06-06-2026 page replacement algorithm

Code:

```
#include<stdio.h>
void FIFO(int pages[],int n,int framecount){
    int frames[framecount];
    int page_fault=0;
    int pointer=0;
    for(int i=0;i<framecount;i++){
        frames[i]=-1;
    }
    printf("FIFO simulation \n REF|\tFRAME\n");
    for(int i=0;i<n;i++){
        int symbol = pages[i];
        int found =0;
        for(int j=0;j<framecount;j++){
            if(symbol==frames[j]){
                found=1;
                break;
            }

        }
        if(!found){
                frames[pointer]=symbol;
                pointer=(pointer+1)%framecount;
                page_fault++;
                printf("%d |",symbol);
                for(int j=0;j<framecount;j++){
                    printf("%d ",frames[j]);
                }
                printf("\n");
            }else{
                printf("%d | HIT \n",symbol);
            }
    }
```

```
    printf("total page faults = %d ",page_fault);


}
int main(){
   int pages[]={1,2,3,2,4,5,6,4};
   FIFO(pages,8,3);
}
```

```
FIFO simulation
 REF|   FRAME
1 |1 -1 -1
2 |1 2 -1
3 |1 2 3
2 | HIT
4 |4 2 3
5 |4 5 3
6 |4 5 6
4 | HIT
total page faults = 6
```

LRU

```
#include<stdio.h>
#include<stdlib.h>
int findLRU(int time[],int n){
    int minimum = time[0],pos=0;
    for(int i=0;i<n;i++){
        if(time[i]<minimum){
            minimum=time[i];
            pos=i;
        }
    }
    return pos;


}
void lru(int pages[],int n,int fcount){
    int frames[fcount];
    int time[fcount];
    int pagefault=0,counter=0;
    for(int i=0;i<fcount;i++){
        frames[i]=-1;
```

```
    time[i]=0;
}
printf("simulation LRU \n");
printf("REF\t|\tFrames\n");
for(int i=0;i<n;i++){
    int symbol = pages[i], frameIndex=-1,found=0;
    for(int j=0;j<fcount;j++){
        if(symbol==frames[j]){
            found=1;
            frameIndex=j;
            break;
        }
    }
    counter++;
    if(!found){
        int replaced =0;
        for(int j=0;j<fcount;j++){
            if(frames[j]==-1){
                frames[j]=symbol;
                time[j]=counter;
                replaced=1;
                pagefault++;
                break;
            }
        }
        if(!replaced){
            int pos = findLRU(time,n);
            frames[pos]=symbol;
            time[pos]=counter;
            pagefault++;
        }
        printf("%d\t| ",symbol);
        for(int j=0;j<fcount;j++){
            printf("%d_",frames[j]);
```

```
            }
            printf("\n");
        }else{
            time[frameIndex]=counter;
            printf("%d  | HIT\n",symbol);
        }
    }
    printf("total fage fault = %d",pagefault);
}
int main(){
    int pages[]={1,2,3,1,2,3,4,5,6};
    lru(pages,9,3);
}
```

```
simulation LRU
REF      |       Frames
1        | 1_-1_-1_
2        | 1_2_-1_
3        | 1_2_3_
1   | HIT
2   | HIT
3   | HIT
4        | 1_2_3_
5        | 1_8_3_
6        | 1_8_9_
total fage fault = 6
```

Optimal

```
#include<stdio.h>
#include<stdlib.h>
void optimal(int pages[],int n,int frameCount){
    int frames[frameCount];
    int pagefault=0;
    printf("\n--- Optimal Simulation ---\nRef | Frames\n");
    for(int i=0;i<frameCount;i++){
        frames[i]=-1;
    }
    for(int i=0;i<n;i++){
        int symbol = pages[i];
        int found=0;
        for(int j=0;j<frameCount;j++){
```

```
    if(symbol==frames[j]){
        found=1;
        break;
    }
}
if(!found){
    int replaced = 0;
    for(int j=0;j<frameCount;j++){
        if(frames[j]==-1){
            frames[j]=symbol;
            pagefault++;
            replaced=1;
            break;
        }
    }
    if(!replaced){
        int fartherest =-1;
        int pos;
        for(int j=0;j<frameCount;j++){
            int nextuse = n;
            for(int k=i+1;k<n;k++){
                if(frames[j]==pages[k]){
                    nextuse=k;
                    break;
                }
            }
            if(nextuse>fartherest){
                fartherest=nextuse;
                pos=j;
            }
        }
        frames[pos]=symbol;
        pagefault++;
    }
```

```
          printf("%d | ",symbol);
          for(int k=0;k<frameCount;k++){
             printf("%d ",frames[k]);
          }
          printf("\n");



       }else{
          printf("%d | HIT",symbol);
       }

    }
    printf("total fage falut = %d",pagefault);
}
int main(){
    int pages[]={7, 0, 1, 2, 0, 3};
    optimal(pages, 6, 3);
}
```

```
--- Optimal Simulation ---
Ref | Frames
7 | 7 -1 -1
0 | 7 0 -1
1 | 7 0 1
2 | 2 0 1
0 | HIT
3 | 3 0 1
total fage falut = 5
```